1.1.d. The cardiovascular and respiratory systems

Learners will develop their knowledge and understanding of the structure and function of the cardiovascular system. Blood vessels and blood cells with their pathway through the heart will be understood along with definitions of key cardiac terms. Learners will understand the pathway of air through the respiratory system and know the role of the respiratory muscles and alveoli during breathing, along with an understanding of key definitions.

Learners will also be able to define aerobic and anaerobic exercise and be able to give practical examples of aerobic and anaerobic activities.

Topic area	Learners must:
Structure and function of the cardiovascular system	• know the double-circulatory system (systemic and pulmonary) • know the different types of blood vessel: ○ arteries ○ capillaries ○ veins • understand the pathway of blood through the heart: ○ atria ○ ventricles ○ bicuspid, tricuspid and semilunar valves ○ septum and major blood vessels: – aorta – pulmonary artery – vena cava – pulmonary vein • know the definitions of: ○ heart rate ○ stroke volume ○ cardiac output • know the role of red blood cells.
Structure and function of the respiratory system	• understand the pathway of air through the respiratory system: ○ mouth ○ nose ○ trachea ○ bronchi ○ bronchiole ○ alveoli • know the role of respiratory muscles in breathing: ○ diaphragm ○ intercostals • know the definitions of: ○ breathing rate ○ tidal volume ○ minute ventilation • understand about alveoli as the site of gas exchange.
Aerobic and anaerobic exercise 	• know the definitions of: ○ aerobic exercise ○ anaerobic exercise • be able to apply practical examples of aerobic and anaerobic activities in relation to intensity and duration.